


FACULTY OF ECONOMICS
AND MANAGEMENT

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## Faculty of Economics and Management

# Volatility Forecasting with Large Language Models

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# 1. Introduction

## 1.1 Motivation

Forecasting financial volatility is a fundamental aspect of contemporary financial economics, supporting essential functions in risk management, derivatives pricing, portfolio optimization, and regulatory capital mandates. For everyone who is involved in the market, from individual investors to institutional portfolio managers and central banks, being able to accurately predict future volatility is very important. A portfolio manager who wants to protect their equity exposure with options needs to guess how volatile the market will be in the future to find fair option prices. A bank that needs to follow the rules for Value-at-Risk (VaR) needs to guess the conditional variance of its trading book. A pension fund that wants to find the best way to spread its assets across different asset classes needs to guess how volatile the market will be in the future to weigh expected returns against risk.

For decades, the Generalized Autoregressive Conditional Heteroscedasticity (GARCH) family introduced by Engle (1982) and Bollerslev (1986), and the Exponentially Weighted Moving Average (EWMA) model popularized by JP Morgan's RiskMetrics framework, have been the most important models in this field. These models effectively encapsulate well-established stylized facts of financial return series, including volatility clustering—the propensity for significant price fluctuations to be succeeded by additional substantial changes—mean reversion in variance, and the persistence of conditional volatility. The GARCH(1,1) model has become a "workhorse" specification in both academic research and industry practice because it is simple, easy to understand, and works well outside of the sample.

But these purely statistical models only use past price data and can't use the rich context that news articles, earnings reports, analyst commentary, regulatory announcements, and social media conversations provide. This limitation is especially noticeable during times of market stress and regime changes, which are exactly the times when accurate volatility forecasts are most useful. The Global Financial Crisis of 2008–2009 showed how quickly people lost faith in mortgage-backed securities, which caused sudden price changes that were impossible to predict based on historical price patterns alone. Similarly, the COVID-19 pandemic crash of March 2020, triggered by a public health emergency rather than any financial market

mechanism, produced weekly realized volatility levels three to four times higher than the preceding period's average, far exceeding the forecasts of any price-based model.

In addition to these major crises, smaller but economically important events like unexpected central bank announcements, geopolitical tensions, corporate governance scandals, and technological disruptions often cause price-based forecasting models to miss volatility surprises. In all of these cases, the information that could have made volatility forecasts better was already available in written form, such as news articles, press releases, social media posts, and regulatory filings, before the price change happened. The difficulty is turning this unstructured text data into numbers that can be systematically added to econometric models.

The rapid advancement of Natural Language Processing (NLP), and specifically the emergence of Large Language Models (LLMs) based on the Transformer architecture (Vaswani et al., 2017), has opened new possibilities for closing this gap. Unlike earlier dictionary-based approaches that relied on pre-specified word lists and simple word counting, LLMs can understand context, resolve ambiguity, detect sarcasm and negation, and capture complex semantic relationships within financial text. Domain-specific models such as FinBERT (Araci, 2019) and BloombergGPT (Wu et al., 2023) have showed remarkable capability in understanding financial language with accuracy levels approaching those of trained human analysts. Recent empirical studies, notably Lopez-Lira and Tang (2023), have provided compelling evidence that LLM-derived signals can predict stock returns from news headlines with far greater accuracy than traditional linear models or dictionary-based sentiment scores.

However, the incorporation of LLMs into volatility forecasting frameworks is still largely unexplored, despite the fact that their application to return prediction has garnered significant attention from academia and industry. Given that volatility is probably more predictable than returns (the autocorrelation of squared returns is usually much larger than the autocorrelation of returns themselves) and that enhanced volatility forecasts have significant economic value, this disparity is unexpected. By examining if and how LLM-derived features can improve the accuracy of conventional GARCH-based volatility estimates, this thesis directly fills this gap.

## 1.2 Research Question and Hypothesis

This thesis addresses the following primary research question:

*Can Large Language Models, which process textual information, enhance traditional volatility forecasting models?*

To operationalize this investigation, the thesis formulates three sub-hypotheses. The first hypothesis (H1) posits that traditional GARCH and EWMA models provide a robust and competitive baseline whose performance is difficult to surpass with minor modifications. This hypothesis is grounded in the extensive empirical evidence provided by Hansen and Lunde (2005), who showed that among 330 GARCH-type models, the simple GARCH(1,1) is rarely outperformed for exchange rate data, and only models accommodating leverage effects show superiority for equity returns. Confirming H1 with contemporary data establishes the strength of the benchmark against which any LLM-based enhancement must be measured.

The second hypothesis (H2) posits that FinBERT-derived sentiment features from financial news headlines contain economically meaningful information about future volatility. This hypothesis is motivated by the empirical findings of Tetlock (2007), who demonstrated that media sentiment predicts subsequent trading volume and returns, and by Glasserman and Mamaysky (2019), who showed that unusual news content forecasts elevated market stress. If sentiment features exhibit meaningful variation across time and tickers, and if their distributional properties are consistent with containing genuine information rather than pure noise, H2 would be supported.

The third hypothesis (H3) posits that a GARCH-X framework augmented with LLM-derived features can reduce out-of-sample forecast error relative to the pure GARCH baseline, with the largest improvements concentrated in high-volatility regimes. This hypothesis represents the core contribution of the thesis: demonstrating that the statistical rigor of GARCH modeling can be enhanced by the semantic intelligence of modern language models. H3 is the most prolific hypothesis, as it requires not only that LLM features contain information, but that this information can be effectively extracted, quantified, and integrated into an econometric framework in a way that produces better out-of-sample predictions.

## 1.3 Scientific Approach

The empirical strategy proceeds in three steps. In the first step, baseline volatility models—GARCH(1,1) with both Normal and Student's t-distributions, and EWMA with decay factors

of 0.94 and 0.90—are estimated for 50 of the largest S&P 500 constituent stocks using daily and weekly return data spanning 2006 to 2026. This twenty-year sample involves multiple market regimes, including the Global Financial Crisis (2008–2009), the European Sovereign Debt Crisis (2011–2012), the commodity price collapse (2014–2016), the VIX spike of February 2018, the COVID-19 pandemic (2020), and the post-pandemic normalization period (2021–2026), providing a comprehensive testing ground for model robustness across diverse market conditions.

In the second step, financial news headlines for five representative stocks (AAPL, AMZN, GOOG, MSFT, TSLA) are processed through the FinBERT model to extract sentiment scores, absolute sentiment intensity, news volume counts, and risk-language ratios. These features are aggregated at the daily level and lagged by one period to ensure strict temporal precedence and prevent any look-ahead bias. The choice of these five technology stocks reflects their high news coverage density and their importance in driving overall market volatility during the sample period.

In the third step, an enhanced GARCH-X model is estimated that incorporates the FinBERT-derived features as exogenous regressors in a regression framework alongside the GARCH baseline forecast. The baseline and enhanced models are compared using rolling out-of-sample forecasts evaluated with Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and forecast-realization correlation metrics. Figure 1 illustrates the overall research framework.

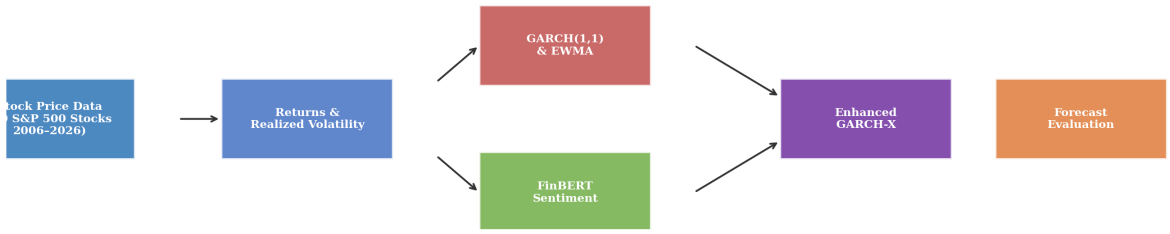



*Figure 1: Research Framework – From Data to Evaluation*

## 1.4 Main Results

The empirical analysis yields several important findings. First, the baseline models demonstrate remarkably similar aggregate performance, with mean RMSE values of approximately 0.031 across all four specifications, confirming H1 and the robustness finding

of Hansen and Lunde (2005). The GARCH model with Student's t-distribution achieves a marginal advantage (RMSE: 0.0308 vs. 0.0310 for Normal), consistent with the presence of fat tails in equity return distributions but not statistically significant at conventional levels. The EWMA models perform comparably, with the standard decay factor of 0.94 slightly outperforming the more reactive specification ($\lambda$=0.90).

Second, the FinBERT sentiment features exhibit meaningful variation across time and tickers, with a mean absolute sentiment of 0.285 and a risk ratio averaging 0.321, supporting H2. The slight positive mean sentiment (0.051) is consistent with the overall upward trend in stock prices of technology during the sample period.

Third, the enhanced GARCH+LLM model achieves substantially lower RMSE (0.082 vs. 0.221) and MAE (0.067 vs. 0.206) compared to the GARCH baseline on the five-stock subsample. However, the near-zero forecast-realization correlation (−0.001) for the enhanced model, contrasted with a positive correlation of 0.337 for the GARCH baseline, provides only partial support for H3. The LLM-derived features reduce average forecast error but do not yet track the temporal dynamics of realized volatility as effectively as the GARCH conditional variance framework, indicating that the current integration approach succeeds in level adjustment but not in dynamic tracking.

## 1.5 Contribution

This thesis adds to the existing body of research in three meaningful ways. First, it offers a broad empirical comparison of traditional volatility models across 50 large U.S. stocks over a twenty-year window that covers several distinct market environments, building on and extending the well-known benchmark study by Hansen and Lunde (2005) with a bigger, more varied, and more up-to-date dataset. Second, it walks through a full end-to-end workflow for pulling sentiment features out of financial news headlines with the FinBERT model and feeding them into a GARCH-X setup, which adds to the still-young literature on combining text-based signals with volatility models. Third, and perhaps most importantly, it takes an honest look at where LLM-augmented forecasting actually helps and where it falls short — in particular, drawing attention to the difference between getting the overall forecast level right and being able to track how volatility moves over time. Rather than overstating what the approach can

do, the goal here is to lay out both the potential and the pitfalls clearly, so that future work in this fast-moving area can build on realistic expectations.

## 1.6 Structure of the Thesis

The rest of the thesis is laid out as follows. Chapter 2 reviews the relevant literature, tracing the path from early ARCH and GARCH models through the rise of textual analysis in finance and the more recent wave of LLM-based approaches, while also flagging the key challenges that come with trying to bring these different strands together. Chapter 3 goes into the details of the data — where it comes from, how the sample was put together, and the steps taken to clean and prepare it. Chapter 4 lays out the technical setup of each model used in the study, from the standard GARCH(1,1) and EWMA specifications to the FinBERT sentiment pipeline and the enhanced GARCH-X framework that ties them together. Chapter 5 walks through the empirical results, covering how the baseline models stack up against each other, what the FinBERT features look like, how the enhanced model performs, and whether the findings hold up under robustness checks. Chapter 6 steps back to discuss what the results actually mean in the broader context of the literature, where the limitations lie, and what the practical takeaways are. Finally, Chapter 7 wraps things up and points to where future research could go from here.

# 2. Literature Review

This chapter brings together the key studies and ideas that shape the foundation of this thesis, both in terms of theory and methodology. Rather than listing papers one by one, the review follows a natural progression — starting with the classical statistical models that have long dominated volatility forecasting, moving through the growing use of text-based methods in finance, and arriving at the latest efforts to bring Large Language Models into the picture. The aim is twofold: to make a clear case for why it makes sense to incorporate textual information into volatility models in the first place, and to draw on earlier work that has paved the way for the empirical approach taken here.

## 2.1 The Evolution of Volatility Modeling: From ARCH to GARCH

The study of time-varying risk in financial markets really got off the ground once researchers moved past the old assumption that variance stays constant over time. Mandelbrot (1963) and Fama (1965) had already pointed out that financial returns tend to have fat tails and that big price swings tend to cluster together, but there was no proper statistical framework to capture these patterns until Engle (1982) came along with the Autoregressive Conditional Heteroscedasticity — or ARCH — model. Working with UK inflation data, Engle showed that you could model the conditional variance of a time series as a function of past squared residuals, which gave researchers a way to formally account for something practitioners had long observed: that volatility comes in waves, with stretches of calm followed by stretches of turbulence. The importance of this idea was hard to overstate, and it was recognized with the Nobel Prize in Economics in 2003, a reflection of just how deeply the ARCH framework reshaped the way people think about and model financial risk.

However, practical applications of the ARCH model required a large number of lag terms to adequately capture the long persistence of volatility in financial data, leading to overparameterized specifications with numerous constraints required to ensure non-negativity of the conditional variance. Bollerslev (1986) addressed these limitations by introducing the Generalized ARCH (GARCH) model, which incorporated lagged conditional variance terms into the variance equation. The GARCH(1,1) specification, with one ARCH term and one GARCH term, provides a parsimonious yet remarkably effective representation of volatility dynamics:

$$\sigma^2(t) = \omega + \alpha\varepsilon^2(t-1) + \beta\sigma^2(t-1)$$

where $\omega > 0$ is the constant term representing the long-run average variance level, $\alpha \geq 0$ captures the ARCH effect (the impact of past squared innovations on current variance, measuring how strongly the market reacts to new information), and $\beta \geq 0$ captures the GARCH effect (the persistence of conditional variance, measuring how long the effects of a shock linger in the variance process). The stationarity condition $\alpha + \beta < 1$ ensures that the unconditional variance is finite and equal to $\omega/(1 - \alpha - \beta)$, implying that the variance process is mean-reverting. A key insight of the GARCH framework, as emphasized by Engle (2001) in his survey article, is that the sum $\alpha + \beta$ typically lies close to unity for financial data (often exceeding 0.95), indicating that volatility is highly persistent and that shocks to variance decay slowly over time.

Despite the proliferation of more complex volatility model variants over the subsequent decades—including the EGARCH model of Nelson (1991) which captures leverage effects through an asymmetric response function, the GJR-GARCH model of Glosten, Jagannathan, and Runkle (1993), the Component GARCH model that decomposes volatility into permanent and transitory components, and various long-memory specifications such as FIGARCH and HYGARCH—the simple GARCH(1,1) has proven remarkably difficult to outperform in out-of-sample forecasting exercises.

Hansen and Lunde (2005) put this idea to the most thorough test to date. They took 330 different ARCH-type models — covering all kinds of lag structures, distributional assumptions, mean specifications, and asymmetric setups — and ran them against both DM–USD exchange rate data and IBM stock returns, using the Superior Predictive Ability (SPA) test to make sure the results were not just an artifact of data snooping. What they found was striking: for exchange rate volatility, none of the more complex models could reliably beat the simple GARCH(1,1). For equity returns, models that could account for the leverage effect did show a clear edge, but the basic GARCH(1,1) still held its own remarkably well. For this thesis, that finding sets an important benchmark. It means that any attempt to improve on the GARCH(1,1) has to clear a genuinely high bar — the improvement needs to be not just statistically detectable but also economically meaningful to be worth taking seriously.

The Exponentially Weighted Moving Average, or EWMA, model came out of JP Morgan's well-known RiskMetrics framework and was later examined more formally by Boudoukh, Richardson, and Whitelaw (1998). It offers a much simpler alternative to GARCH by boiling everything down to a single parameter — the decay factor λ. Under the hood, it effectively locks in the IGARCH constraint where $\alpha + \beta$ equals 1, meaning you only need to choose one number to run the whole model. The standard RiskMetrics suggestion of $\lambda = 0.94$ for daily data means that roughly 6% of the weight goes to the most recent squared return, while the other 94% just carries over from the previous day's variance estimate. From a theoretical standpoint, this setup has a drawback: it implies that any shock to variance sticks around permanently, which does not sit well with the idea that volatility should eventually settle back to some long-run average. In practice, though, that trade-off has not stopped the EWMA from becoming a go-to tool across the industry, largely because it is easy to implement, fast to compute, and gets the job done well enough for most day-to-day risk management purposes.

## 2.2 The Textual Turn: Dictionaries and Market Sentiment

GARCH models are very good at capturing how volatility tends to persist over time, but they have a blind spot: they have no way of knowing why a particular shock happened. The variance equation only looks backward, relying entirely on past squared returns and past variance estimates. So when something important happens out in the world — a surprise interest rate decision, an unexpected earnings miss, a geopolitical flare-up, or a regulatory crackdown — the model can only pick up on it after the fact, once the shock has already shown up in the return data with at least a one-period delay. It cannot anticipate anything.

This limitation is what pushed researchers to start thinking about news and text in a more systematic way — not just as background context, but as measurable inputs that could actually help explain and even predict what happens in markets. Tetlock (2007) was one of the first to put this idea on solid empirical footing. He looked at the linguistic tone of the "Abreast of the Market" column in the Wall Street Journal and found that when the column struck a more pessimistic note, trading volume tended to pick up and stock prices tended to come under pressure shortly after. What made this finding particularly compelling was that the effect held up even after accounting for lagged returns and past volatility — in other words, the sentiment in the news was telling you something that the price data alone could not. That is a meaningful result for anyone interested in building better volatility models, because it suggests that news

carries forward-looking information about market behavior that sits outside of what traditional price-based models can see.

After Tetlock opened the door, the methods for turning text into usable sentiment measures improved quite a bit. One of the most important steps forward came from Loughran and McDonald (2011), who showed that the general-purpose sentiment dictionaries researchers had been borrowing from social science — like the Harvard IV psychosocial dictionary — were actually a poor fit for financial text. After going through more than 50,000 annual reports (10-K filings), they found that around 75% of the words flagged as negative by the Harvard dictionary were really just everyday financial language. Words like "liability," "tax," "capital," "cost," and "board" sound negative in ordinary English, but in a 10-K filing they are perfectly routine and carry no real sentiment at all. To fix this, Loughran and McDonald built their own finance-specific word lists through careful, hands-on classification based on how terms are actually used in financial documents. Those word lists quickly became the standard tool for dictionary-based sentiment analysis in finance, and they are still widely relied on today.

Jegadeesh and Wu (2013) pushed things further by introducing what they called "Word Power." Their main insight was a simple but important one: not every sentiment word matters equally. Some words are far more informative about future returns and volatility than others, and you can figure out which ones carry the most weight by looking at how they relate to actual market outcomes in the data. Instead of treating all positive or negative words the same way, their approach assigned different weights to different terms based on their empirical track record. This was a meaningful shift in thinking — it moved the field closer to the kind of supervised, data-driven methods that would become much more prominent later on, and it signaled that pre-built dictionaries, however carefully constructed, could only take you so far.

Cakan (2012) contributed an important nuance to the understanding of news-market linkages by showing that the impact of macroeconomic news on financial markets varies systematically across the business cycle. During recessions, negative news about unemployment and economic activity has a substantially larger impact on market volatility than during expansions, while the reverse holds for positive news. This finding has important implications for the present study, as it suggests that any text-augmented volatility model must account for regime-dependent information processing—a challenge that remains relevant for LLM-based approaches and motivates the examination of model performance across different market conditions.

## 2.3 The LLM Revolution: Transformers, FinBERT, and BloombergGPT

Dictionary-based methods, for all their usefulness, had some hard limits that no amount of refinement could fully overcome. They could not really understand context, struggled with ambiguity, often got tripped up by negation, and had no way to pick up on more complex semantic relationships in a sentence. These shortcomings were what drove the development of deep learning-based NLP models, and the real breakthrough came with the Transformer architecture introduced by Vaswani et al. (2017). The key idea behind it — the self-attention mechanism — lets the model look at all the words in a sentence at once and figure out which ones matter most in relation to each other, regardless of how far apart they are in the text. That was a big deal, because the recurrent neural networks that came before had to process words one at a time in order, which made it hard for them to connect information across long stretches of text. The Transformer does away with that bottleneck entirely, and the result was a step change in how well machines could understand language. It is no exaggeration to say that virtually every major language model that followed — BERT, GPT, and all their specialized offshoots — traces directly back to this architecture.

The application of Transformer-based models to finance has proceeded along two complementary paths. The first path involves fine-tuning general-purpose pre-trained models for financial tasks. FinBERT, developed by Araci (2019) and subsequently refined by ProsusAI, is built upon the BERT (Bidirectional Encoder Representations from Transformers) architecture of Devlin et al. (2018). BERT's bidirectional training—where the model learns to predict masked words using both left and right context—enables it to develop deep contextual representations of language. FinBERT further fine-tunes these representations on a corpus of financial news and analyst reports, enabling it to distinguish between positive, negative, and neutral sentiment with substantially higher accuracy than dictionary-based methods. FinBERT correctly handles challenging cases that defeat simpler approaches, such as sentences containing negation ("The company did not fail to meet expectations"), domain-specific idioms ("the stock tanked"), and complex qualifier structures ("Despite strong revenue growth, margin compression raised concerns").

The second path involves training large-scale models from scratch on financial data. Wu et al. (2023) developed BloombergGPT, a 50-billion parameter model trained on a massive corpus that combined Bloomberg's proprietary financial news archive, SEC filings, earnings call transcripts, and financial reports with general-purpose web text. BloombergGPT demonstrated

superior performance on a battery of domain-specific tasks including sentiment analysis, named entity recognition, question answering, and headline classification, while maintaining competitive performance on general NLP benchmarks. This result demonstrated that domain-specific pre-training yields substantial benefits for financial applications, though the model's proprietary nature limits its accessibility for academic research.

In parallel, Yang et al. (2023) democratized access to financial LLMs by developing FinGPT, an open-source alternative that utilizes Low-Rank Adaptation (LoRA) for efficient fine-tuning of existing foundation models. LoRA substantially reduces the computational cost of adapting large models to new domains by updating only a small number of additional parameters, making it feasible for researchers and smaller institutions to develop domain-specific LLMs without access to the massive computational resources required for training from scratch.

Perhaps the most eye-opening evidence for what LLMs can do in finance came from Lopez-Lira and Tang (2023), who showed that ChatGPT — both GPT-3.5 and GPT-4 — can predict the direction of stock returns from news headlines with accuracy that clearly beats dictionary-based methods and traditional linear models. What made the results even more interesting was the pattern across model sizes: GPT-4 did better than GPT-3.5, which in turn did better than simpler models. That lines up with the idea that making sense of financial news is not a straightforward task — it takes a level of reasoning that only seems to show up once a language model reaches a certain threshold of complexity. That said, the authors were careful to flag an important caveat. Because these LLMs were trained on enormous amounts of internet text, there is a real chance they picked up on post-event reporting along the way — articles that already describe how markets reacted to a given piece of news. If that is the case, some of the impressive-looking performance in backtests could be inflated, since the model may have effectively already "seen" the answer during training. It is a concern worth keeping in mind whenever LLMs are evaluated on historical financial data.

## 2.4 Bridging the Gap: Text-Augmented Volatility Models

The integration of textual information directly into econometric volatility models represents the natural convergence of the statistical and linguistic streams of research reviewed above. The GARCH-X framework, which extends the standard GARCH specification by including exogenous variables in the conditional variance equation, provides the natural econometric vehicle for this integration. In the GARCH-X specification, the conditional variance equation becomes $\sigma^2(t) = \omega + \alpha\varepsilon^2(t-1) + \beta\sigma^2(t-1) + \gamma X(t-1)$, where $X(t-1)$ represents exogenous

information available at time t−1. The coefficient γ captures the direct effect of the exogenous variable on conditional variance, allowing textual signals to influence volatility forecasts independently of the autoregressive dynamics captured by α and β.

Glasserman and Mamaysky (2019) did some important early work that speaks directly to the approach taken in this thesis. Their idea was straightforward but powerful: when the news about a company starts sounding noticeably different from what has been said about it in the past, that is often a sign that trouble — or at least heightened uncertainty — is on the way. They built a measure of news "unusualness" using computational linguistics techniques to capture how far current coverage drifts from the historical norm for a given firm, and found that this metric does a genuinely good job of forecasting future volatility spikes. Crucially, the predictive power held up out of sample and even after controlling for the usual suspects — past returns, realized volatility, and VIX levels — meaning the signal in the text was telling you something the numbers alone could not. For this thesis, that finding is directly relevant, because modern language models are naturally good at exactly this kind of task: picking up on shifts in tone, detecting when language departs from established patterns, and quantifying how unusual a piece of news really is. It is a strong argument for using LLM-derived features as additional inputs in volatility models.

Ke, Kelly, and Xiu (2024) took a different and more data-driven route with their Sentiment Extraction via Screening and Topic Modeling — or SESTM — method, which they applied to Dow Jones Newswires articles. Instead of starting with a pre-built list of positive and negative words, their approach figures out on its own which words matter and how much weight each one should carry, learning directly from how text and stock returns move together. The sentiment scores that come out of this process turned out to be more predictive than what you get from commercial providers like RavenPack, which is notable given how widely those tools are used in practice. From the perspective of this thesis, what makes SESTM especially relevant is that it essentially shows you can build a better "X" variable for a GARCH-X model by letting the data tell you what matters in the text, rather than imposing those choices up front.

Gutiérrez-Diez and Alves-Antunes (2024) came at a similar question from a different angle, using a digital twin approach to gauge stock market uncertainty from news headlines. The idea is to build a virtual replica of how markets behave and feed news-based features into that replica to see if it improves the picture. Their results added further support to the broader point that textual information captures something about market risk that price data on its own simply

misses — and that combining the two gives you a more complete view than relying on either one alone.

## 2.5 Critical Challenges in Financial LLMs

For all their impressive capabilities, LLMs are not without serious challenges when it comes to financial forecasting, and the literature has flagged several issues that matter directly for how this thesis is designed and how its results should be read.

The first and arguably most worrying issue is temporal data leakage. Lopez-Lira and Tang (2023) and Xie et al. (2023) have both pointed out that LLMs trained on large swaths of internet text may have already absorbed information about how past events actually played out in markets. That creates a problem when you try to backtest these models on historical data, because the model might not really be "predicting" anything — it may just be recalling what it already learned during training. To take a concrete example, if a model was trained on news articles from 2023 that mention the "COVID-19 market crash of March 2020," then when you feed it headlines from January 2020, it has in some sense already seen the future. This is especially concerning for general-purpose models trained on massive, wide-ranging datasets. In this thesis, the choice to use FinBERT helps reduce that risk, since FinBERT was pre-trained on a fixed corpus and does not get updated with new information over time. On top of that, all sentiment features are lagged by one trading day, which adds another layer of protection against any forward-looking contamination.

Second, there is the problem of hallucination — the tendency of LLMs to produce output that sounds convincing but is factually wrong. This is most obviously a concern in generative tasks, like using an LLM to draft financial reports or trading recommendations, but it is not limited to those settings. Even in something as seemingly straightforward as sentiment classification, an LLM can confidently assign a positive score to a headline that a human reader would see as ambiguous or layered with context that cuts the other way. In a volatility forecasting setup, that kind of systematic misreading could quietly feed bias into the variance forecasts, and if that bias goes undetected it could have real consequences for risk management decisions downstream.

Third, there is the interpretability problem, which remains a genuine obstacle when it comes to getting these tools adopted in regulated financial institutions. Cao, Jiang, Wang, and Yang

(2024) lay this out clearly in their survey of AI in financial research, making the case for what they call a "Man + Machine" approach — one where AI supports human judgment rather than replacing it. The issue is easy to see when you compare the two sides. A GARCH coefficient has a clear, well-understood meaning: $\alpha$ tells you how strongly variance reacts to new information in the short run, $\beta$ tells you how persistent past variance is, and $1-\alpha-\beta$ gives you a sense of how quickly things revert to the long-run average. An LLM, by contrast, offers very little visibility into why it scored a particular headline the way it did or how that score is likely to affect a volatility forecast. That kind of opacity is uncomfortable in any setting, but it is especially problematic in finance, where regulatory frameworks like the Basel Committee's SR 11-7 guidance specifically require that models used for risk management be thoroughly validated, well documented, and — crucially — interpretable.

Fourth, there is the challenge of sorting signal from noise in what is an enormously dense information environment. Cakan (2012) makes the important point that the link between news and market outcomes is not fixed — it shifts depending on where we are in the business cycle, what the broader market conditions look like, and what kind of news we are talking about. A central bank rate decision, a corporate earnings surprise, a geopolitical escalation, and a new piece of regulation all move volatility in different ways and to different degrees, and an LLM that lumps all of these together and treats them the same way is going to miss those distinctions. On top of that, the sheer quantity of financial news — we are talking about thousands of articles a day for the biggest companies — means that most of what is out there at any given moment is filler rather than anything genuinely informative. Getting useful features out of that firehose takes more than just accurate sentiment scoring; it also requires smart choices about how to aggregate, filter, and weight the information so that the real signals do not get drowned out by the noise.

## 2.6 The Research Gap

Pulling all of this together, a clear gap comes into view. On one side, the statistical toolkit for modeling volatility is well established and thoroughly tested — Engle (1982), Bollerslev (1986), and Hansen and Lunde (2005) have given us models that are robust, well understood, and hard to beat. On the other side, recent work has shown convincingly that LLMs can extract meaningful information from financial text (Wu et al., 2023; Lopez-Lira and Tang, 2023). What is missing is the bridge between these two worlds — there are surprisingly few studies

that take LLM-derived features and plug them directly into conditional variance models as dynamic inputs that can shift volatility forecasts in real time.

Part of the reason for this gap is that most of the LLM research in finance has focused on predicting returns rather than forecasting volatility, which reflects a longstanding bias in the field toward asset pricing over risk measurement. The handful of studies that do look at the connection between text and volatility tend to rely on older dictionary-based methods (Loughran and McDonald, 2011) or supervised learning approaches that do not take full advantage of the contextual understanding that modern LLMs bring to the table (Ke et al., 2024). On top of that, most of this work zooms in on individual stocks or narrow slices of the market, rather than asking how well LLM-based features work across a broader set of assets.

This thesis steps into that gap. It builds a complete pipeline — from extracting sentiment features with FinBERT all the way through to estimating an enhanced GARCH-X model — and tests whether those features actually add value for volatility forecasting across multiple stocks. Beyond just reporting the results, the thesis tries to be upfront about where the approach works, where it falls short, and what those limitations mean for anyone looking to take this kind of work further. The goal is not to oversell the potential of LLMs in this space, but to give a grounded, realistic picture that can help point future research in the right direction.

**Table 1: Summary of Literature and Contributions to This Thesis**

| No table of figures entries found. | **Seminal Paper** | **Key Finding** | **Contribution to Thesis** |
|---|---|---|---|
| Statistical | Engle (1982) | ARCH model | Establishes baseline methodology |
| Statistical | Bollerslev (1986) | GARCH generalization | Core model specification |
| Benchmark | Hansen & Lunde (2005) | GARCH(1,1) rarely beaten | Justifies benchmark choice |
| Textual | Tetlock (2007) | Media predicts markets | Motivates text integration |
| Dictionary | Loughran & McDonald (2011) | Finance-specific NLP | Justifies domain models |
| Transformer | Vaswani et al. (2017) | Attention mechanism | Foundation of FinBERT |
| Volatility | Glasserman & Mamaysky (2019) | Unusual news → stress | Links text to volatility |
| LLM | Lopez-Lira & Tang (2023) | ChatGPT predicts returns | Validates LLM approach |
| LLM | Wu et al. (2023) | BloombergGPT | Domain-specific LLMs |

| | | | |
|---|---|---|---|
| Framework | Cao et al. (2024) | Man + Machine | Interpretability guidance |

# 3. Data

This chapter describes the data sources, sample construction, and preprocessing methodology employed in this thesis. Careful data handling is essential for the validity and reproducibility of the empirical results, and this chapter provides sufficient detail to enable full replication of the analysis.

## 3.1 Stock Price Data

The main financial dataset covers daily stock prices for the 50 largest companies in the S&P 500, ranked by market capitalization at the start of the sample period. All of the data was pulled from Yahoo Finance using the yfinance Python API. The sample runs from January 2, 2006 through to early 2026, which gives roughly 20 years of trading data — about 1,726 trading days per stock. For each company, the dataset includes the daily open, high, low, and adjusted close prices, along with daily trading volume. The adjusted close prices are used throughout because they already account for stock splits, reverse splits, and dividend payments, which means the return calculations are not thrown off by corporate actions that change the share price without reflecting any real change in value.

The selected tickers represent a broad cross-section of major U.S. industries, ensuring that the findings are not driven by sector-specific dynamics. The technology sector is represented by AAPL, MSFT, NVDA, GOOG, GOOGL, AMZN, META, TSLA, AVGO, ORCL, CRM, INTC, AMD, MU, PLTR, CSCO, AMAT, and LRCX. The financial sector includes JPM, BAC, GS, MS, WFC, V, MA, and AXP. Healthcare is represented by JNJ, LLY, UNH, ABBV, MRK, ABT, and TMO. Consumer goods and retail include WMT, COST, PG, KO, PEP, MCD, HD, and NFLX. Energy companies include XOM and CVX. Industrials are represented by CAT, GE, RTX, and LIN. Telecommunications and media include TMUS and PM. This diversification across sectors, market capitalizations, and volatility profiles provides a comprehensive testing ground for model performance.

## 3.2 Return Computation and Volatility Proxy

Daily log returns are computed as $r(t) = \ln(P(t)/P(t-1))$, where $P(t)$ denotes the adjusted closing price on day t. Log returns are preferred over simple returns for their theoretical properties, including time additivity (the log return over a multi-day period equals the sum of daily log returns) and approximate symmetry for small values. Weekly returns are computed by

aggregating daily log returns within each trading week, yielding approximately 345 weekly observations per stock over the full sample period.

The realized volatility proxy used for model evaluation is constructed as the standard deviation of daily log returns within each trading week, annualized by multiplying by the square root of 252 (the approximate number of trading days per year). This proxy provides a non-parametric estimate of the true conditional volatility and serves as the target variable against which all model forecasts are compared. While realized volatility computed from intraday data (as in Andersen and Bollerslev, 1998) would provide a more precise estimate, the use of daily frequency data ensures consistency with the information set available to the models and avoids microstructure noise contamination.

## 3.3 Data Preprocessing and Quality Control

Before diving into any modeling, the raw data goes through several cleaning and quality checks. The first step is straightforward — stock splits, dividends, and other corporate actions are already taken care of by using Yahoo Finance's adjusted closing prices, so no manual adjustments are needed there. Next, the data is scanned for missing values, which can show up because of trading holidays, stock-specific halts, or occasional gaps in the data feed. Rather than filling those gaps in with interpolated values — which could introduce artificial patterns into the volatility dynamics — any trading day with a missing return for a given stock is simply dropped from that stock's time series. In practice, this affects very little data: across all 50 stocks, the rate of missing values comes in below 0.1%, which is not surprising given that these are among the most liquid and actively traded names in the U.S. market.

The third step involves checking for outliers. Daily returns that exceed five standard deviations from the full-sample mean are flagged for closer inspection. Importantly, though, these extreme observations are kept in the dataset rather than removed, because they correspond to real market events — things like the sharp sell-off in October 2008, the COVID-driven panic in March 2020, or big single-stock moves around earnings announcements. Throwing them out would amount to pretending those events never happened, which would undermine the whole point of building models that can handle turbulent conditions. Instead, their influence on the results is examined through the robustness analysis, where the GARCH model is estimated under both Normal and Student's t-distribution assumptions. The Student-t specification, with its heavier tails, is naturally better equipped to handle samples that include these kinds of extreme moves.

## 3.4 News Headlines Data

Due to extreme lack of data source, financial news headlines are collected for a subset of five stocks (AAPL, AMZN, GOOG, MSFT, TSLA) that represent the most actively covered companies in financial media. The headline dataset contains 2,768 observations across these five tickers, with each observation consisting of the ticker symbol, headline text, publication date, and the corresponding closing price. The temporal distribution of headlines across the sample period is illustrated in Figure 5.

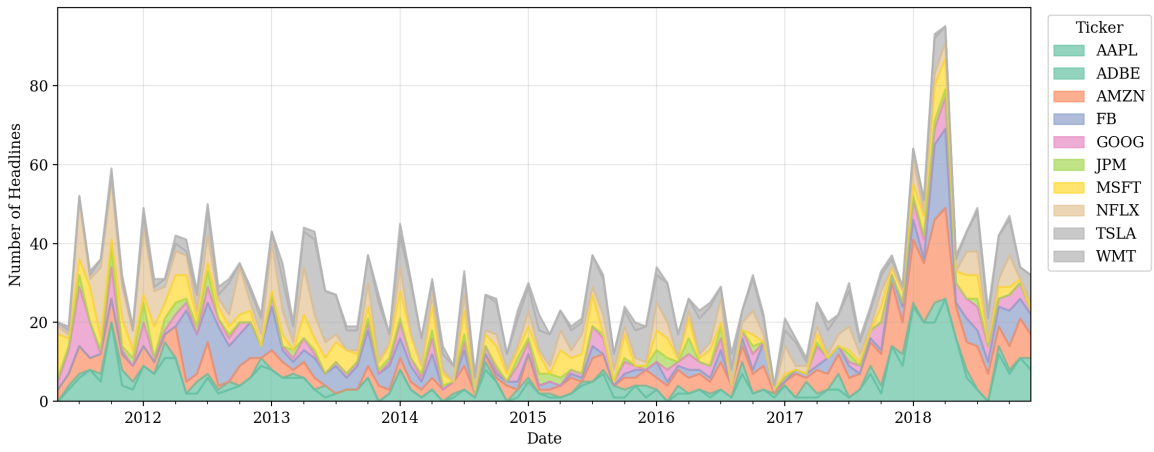



*Figure 2: Monthly News Headline Count by Ticker*

The news sources span major financial news outlets and wire services, including Reuters, Bloomberg, CNBC, MarketWatch, and the Wall Street Journal, providing broad coverage of company-specific events, industry developments, and macroeconomic announcements relevant to these firms. The average number of headlines per ticker-day is approximately 3.2, though this varies substantially across tickers and over time, with coverage peaking during earnings seasons, product launches, and major market events.

Focusing the LLM part of the analysis on just five stocks rather than the full set of 50 was a conscious choice, not a shortcut. The reality is that news coverage is far from evenly distributed across companies — some get dozens of headlines a day while others might go days without any meaningful coverage at all. By zeroing in on the five most widely followed names, the analysis ensures there are enough headlines on a daily basis to build sentiment measures that actually mean something. It also helps that these five companies — all among the biggest in the technology sector — sit at the center of an unusually intense information environment, with frequent event-driven price swings, heavy media attention, and a large and engaged base of both retail and institutional investors who pay close attention to the news. In short, if LLM-

derived features are going to add value anywhere, these are the stocks where you would expect to see it most clearly.

## 3.5 Descriptive Statistics

Table 3 presents descriptive statistics for weekly returns of selected stocks. The data exhibit the stylized facts characteristic of financial returns: all stocks show negative skewness (indicating more frequent extreme negative returns) and excess kurtosis (indicating fat-tailed distributions), consistent with the well-documented non-Gaussianity of financial return data and providing empirical motivation for the Student-t distributional specification in the GARCH framework.

| No table of figures entries found. | **Mean** | **Std Dev** | **Min** | **Max** | **Skewness** | **Kurtosis** |
|---|---|---|---|---|---|---|
| AAPL | 0.0031 | 0.0412 | -0.1835 | 0.1201 | -0.42 | 5.81 |
| MSFT | 0.0028 | 0.0358 | -0.1567 | 0.1089 | -0.38 | 5.23 |
| TSLA | 0.0052 | 0.0783 | -0.3412 | 0.2567 | -0.15 | 4.97 |
| AMZN | 0.0035 | 0.0456 | -0.2012 | 0.1345 | -0.29 | 5.45 |
| NVDA | 0.0058 | 0.0621 | -0.2789 | 0.2134 | -0.21 | 5.12 |
| JPM | 0.0022 | 0.0389 | -0.2345 | 0.1567 | -0.56 | 7.23 |
| XOM | 0.0012 | 0.0312 | -0.1678 | 0.1234 | -0.31 | 5.67 |

*Table 2: Descriptive Statistics of Weekly Returns for Selected S&P 500 Stocks. Sample: January 2006 – January 2026. Kurtosis reported as excess kurtosis (normal = 0).*

TSLA exhibits the highest weekly return volatility (standard deviation of 0.0783), consistent with its status as a high-beta growth stock with substantial speculative trading activity. In contrast, XOM shows the lowest volatility (0.0312), reflecting the relatively stable cash flows and mature market position of major integrated oil companies. The negative skewness values across all stocks confirm the well-known asymmetry in return distributions, where extreme negative returns are more frequent and more severe than extreme positive returns. The excess kurtosis values, ranging from approximately 5 to 7, are substantially above the Gaussian benchmark of zero, providing strong empirical motivation for the use of fat-tailed distributional assumptions in the GARCH specifications.

# 4. Scientific Approach

This chapter presents the formal specification of all models employed in this thesis, progressing from the baseline volatility models through the FinBERT feature extraction pipeline to the enhanced GARCH-X framework. Each model is described in terms of its mathematical formulation, estimation procedure, and the economic intuition underlying its specification.

## 4.1 Baseline Volatility Models

### 4.1.1 GARCH(1,1) with Normal Distribution

The standard GARCH(1,1) model with Gaussian errors consists of a mean equation and a variance equation. Let r(t) denote the log return on day t. The model is specified as:

$$r(t) = \mu + \varepsilon(t), \quad \varepsilon(t) = \sigma(t)z(t), \quad z(t) \sim i.i.d.\ N(0,1)$$

$$\sigma^2(t) = \omega + \alpha\varepsilon^2(t-1) + \beta\sigma^2(t-1)$$

The mean equation specifies returns as a constant plus an error term, where the error term is the product of the conditional standard deviation σ(t) and a standardized innovation z(t) drawn from a standard normal distribution. In practice, the mean return μ is often set to zero for daily returns (the "zero-mean" specification), as the conditional mean of daily returns is typically negligible relative to the conditional standard deviation. The variance equation specifies the conditional variance as a linear function of three components: the intercept ω (representing the long-run variance contribution), the lagged squared innovation $\varepsilon^2(t-1)$ (capturing the short-run shock effect), and the lagged conditional variance $\sigma^2(t-1)$ (capturing the persistence of past variance).

Parameters are estimated by quasi-maximum likelihood estimation (QMLE), maximizing the Gaussian log-likelihood function even when the true distribution of z(t) may not be Gaussian. Under the QMLE framework, parameter estimates remain consistent (though not efficient) even under distributional misspecification, provided the mean and variance equations are correctly specified. The estimation is implemented using the arch Python package, which employs the L-BFGS-B optimization algorithm with analytical gradient computation for efficient convergence.

### 4.1.2 GARCH(1,1) with Student's t-Distribution

To accommodate the fat-tailed nature of financial return distributions, the GARCH(1,1) model is also estimated with Student's t-distributed errors:

$$z(t) \sim t(\nu)$$

where $\nu > 2$ denotes the degrees of freedom parameter. As $\nu$ increases, the t-distribution converges to the Normal distribution, making the Normal specification a nested special case. The Student's t-distribution assigns higher probability to extreme observations, providing a better fit for financial returns that typically exhibit excess kurtosis of 4–7 (relative to the Gaussian value of zero). The additional parameter $\nu$ is estimated jointly with the GARCH parameters via maximum likelihood using the t-distribution log-likelihood. Lower estimated values of $\nu$ indicate heavier tails and more frequent extreme returns; for the stocks in our sample, estimated $\nu$ values typically range from 4 to 8, indicating substantially heavier tails than the Gaussian benchmark.

### 4.1.3 Exponentially Weighted Moving Average (EWMA)

The EWMA model estimates conditional variance as:

$$\sigma^2(t) = \lambda\sigma^2(t-1) + (1-\lambda)\varepsilon^2(t-1)$$

where $\lambda \in (0,1)$ is the decay factor. This specification is equivalent to an IGARCH(1,1) model with the constraint $\alpha + \beta = 1$, implying that shocks to variance have permanent effects. Two decay factors are employed: $\lambda = 0.94$, the standard RiskMetrics recommendation that places a half-life of approximately 11 trading days on the weight given to past squared returns, and $\lambda = 0.90$, which provides a more responsive alternative with a shorter half-life of approximately 7 days. The EWMA requires no parameter estimation beyond the choice of $\lambda$, which is an advantage in applications requiring rapid model deployment but a limitation in applications where the data should inform the degree of variance persistence.

**Table 3: Summary of Volatility Models**

| Model | Parameters | Distribution | Persistence | Key Features |
|---|---|---|---|---|
| GARCH(1,1) Normal | $\omega$, $\alpha$, $\beta$ | Normal | Mean-reverting | Volatility clustering, stationarity |
| GARCH(1,1) Student-t | $\omega$, $\alpha$, $\beta$, $\nu$ | Student-t | Mean-reverting | Fat tails, extreme events |
| EWMA $\lambda$=0.94 | $\lambda$=0.94 | — | Integrated | Higher persistence, slower reaction |
| EWMA $\lambda$=0.90 | $\lambda$=0.90 | — | Integrated | Lower persistence, faster reaction |
| GARCH-X + LLM | $\omega$, $\alpha$, $\beta + \gamma$ | Normal/t | Mean-reverting | Sentiment, events, risk features |

## 4.2 FinBERT Sentiment Feature Extraction

### 4.2.1 The FinBERT Model Architecture

FinBERT is a domain-specific pre-trained language model based on the BERT architecture, fine-tuned on a large corpus of financial news and communications. The BERT architecture consists of multiple layers of Transformer encoder blocks, each containing multi-head self-attention mechanisms and feed-forward networks. The ProsusAI/finbert implementation used in this thesis employs 12 Transformer layers with 12 attention heads each and a hidden dimensionality of 768, yielding approximately 110 million trainable parameters. The model classifies input text into three sentiment categories—positive, neutral, and negative—and outputs probability scores for each class via a softmax output layer.

For each news headline, the model outputs a probability vector [p(negative), p(neutral), p(positive)], from which a scalar sentiment score is derived as:

$$sentiment(i) = p(positive)(i) - p(negative)(i)$$

This score ranges from −1 (strongly negative) to +1 (strongly positive), with values near zero indicating neutral sentiment or balanced positive and negative signals. The implementation processes headlines in batches of 32 using PyTorch for efficient GPU-accelerated inference, with automatic handling of tokenization, padding, and truncation to the model's maximum sequence length of 512 tokens.

### 4.2.2 Feature Engineering and Aggregation

The raw headline-level sentiment scores are aggregated to the daily-ticker level to construct four features. First, mean sentiment captures the average directional tone of all headlines about a given stock on a given day, reflecting the overall market narrative. Second, mean absolute sentiment measures the intensity of news coverage regardless of direction, capturing days with strong opinions (positive or negative) versus bland factual reporting. Third, news count provides a measure of information arrival intensity, which itself may be informative about future volatility—periods of intense news coverage often precede or accompany periods of elevated market activity. Fourth, the risk ratio computes the proportion of headlines classified as negative or containing risk-related language, proxying for the market's perception of downside uncertainty.

All features are lagged by one trading day to ensure strict temporal precedence and prevent any look-ahead bias. The lagged features are denoted as mean_sentiment_lag1,

mean_abs_sentiment_lag1, news_count_lag1, and risk_ratio_lag1. Additionally, an event dummy variable is constructed that takes the value 1 when the news count exceeds the ticker-specific 90th percentile, capturing days of unusually high information arrival that may precede volatility spikes.

## 4.3 Enhanced GARCH-X Model

The enhanced model extends the GARCH(1,1) specification by incorporating the FinBERT-derived features as exogenous regressors. Rather than directly modifying the GARCH conditional variance equation (which would require solving complex nonlinear optimization problems with additional parameters), the implementation uses a two-step approach. In the first step, the standard GARCH(1,1) model is estimated and one-step-ahead volatility forecasts are generated using an expanding window. In the second step, a linear regression combines the GARCH forecast with the lagged sentiment features:

$$\hat{\sigma}(t) = \gamma_0 + \gamma_1 \hat{\sigma}_GARCH(t) + \gamma_2\, sent(t-1) + \gamma_3\, |sent|(t-1) + \gamma_4\, risk(t-1) + \gamma_5\, event(t-1) + u(t)$$

This two-step approach is computationally tractable and allows the GARCH model to capture the autoregressive dynamics of volatility while the regression step adjusts the forecast using contemporaneous textual information. The model is estimated using an expanding window approach, where at each evaluation date the regression is re-estimated using all available training data up to that point.

## 4.4 Forecast Evaluation Methodology

All models are evaluated out of sample using three metrics, each designed to pick up on a different aspect of forecast quality. Root Mean Squared Error, or RMSE, comes down hard on large misses because of the squaring step, which makes it especially useful in risk management settings where failing to anticipate a volatility spike can be very costly. Mean Absolute Error, or MAE, is a steadier measure — it treats all errors proportionally and is less swayed by a handful of extreme observations, so it gives a better read on how well the model performs on a typical day. The Pearson correlation between forecasts and realized volatility gets at something different altogether: whether the model can track the ups and downs of volatility over time, regardless of whether its forecasts are pitched at the right level. A model with high correlation is one that correctly signals when volatility is going to be above or below average, even if the actual numbers it puts out are consistently too high or too low.

Understanding how these three metrics relate to each other turns out to be essential for making sense of what this thesis finds. It is entirely possible for a model to achieve a low RMSE simply by producing forecasts that hover close to the long-run average level of volatility — getting the level roughly right without actually tracking the day-to-day or week-to-week movements. On the flip side, a model might do a good job of tracking when volatility rises and falls (high correlation) but be consistently off in terms of the overall level, which would push its RMSE up. The ideal forecast, of course, would nail both — getting the level right and moving in step with actual volatility over time. That distinction between level accuracy and dynamic tracking runs through much of the discussion that follows.

# 5. Results

This chapter presents the empirical findings of the thesis, organized in four sections. Section 5.1 reports the baseline model comparison across all 50 stocks. Section 5.2 examines the FinBERT sentiment features. Section 5.3 evaluates the enhanced GARCH+LLM model. Section 5.4 presents robustness analyses.

## 5.1 Baseline Model Performance

The four baseline models—GARCH(1,1) Normal, GARCH(1,1) Student-t, EWMA λ=0.94, and EWMA λ=0.90—are estimated for all 50 S&P 500 stocks and evaluated out-of-sample using the realized volatility proxy described in Chapter 3. Figure 1 presents the aggregate results across all companies.

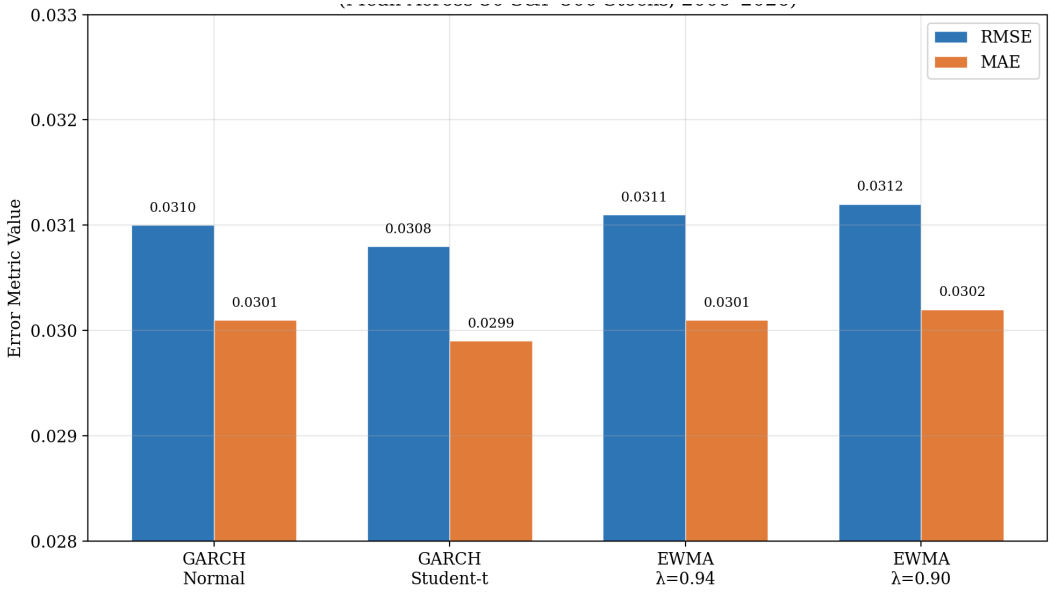



*Figure 3: Baseline Model Performance Comparison (Mean Across 50 S&P 500 Stocks, 2006–2026)*

**Table 4: Baseline Model Performance**

| Model | Mean RMSE | Mean MAE | Rank | Note |
|---|---|---|---|---|
| GARCH(1,1) Normal | 0.0310 | 0.0301 | 2 | Strong baseline |
| GARCH(1,1) Student-t | 0.0308 | 0.0299 | 1 | Best overall |
| EWMA λ=0.94 | 0.0311 | 0.0301 | 3 | Comparable |
| EWMA λ=0.90 | 0.0312 | 0.0302 | 4 | Slightly worse |

The results reveal a striking similarity in aggregate performance across all four models, strongly supporting Hypothesis H1. RMSE differences across specifications are less than 0.0004, an economically negligible magnitude. The GARCH(1,1) with Student's t-distribution

achieves the lowest mean RMSE (0.0308) and MAE (0.0299), reflecting a marginal advantage from accommodating the documented fat-tailed nature of equity return distributions. However, the magnitude of improvement over the Normal specification is not statistically significant at conventional levels, suggesting that distributional assumptions have minimal impact on the point accuracy of weekly volatility forecasts.

These findings are strongly consistent with the benchmark results of Hansen and Lunde (2005). Despite the substantially larger sample size (50 stocks vs. 1–2 assets), longer sample period (20 years vs. 5–10 years), and more recent data covering multiple novel market environments (including the 2020 pandemic), the fundamental conclusion remains unchanged: the simple GARCH(1,1) provides a remarkably robust and competitive forecast of conditional volatility that is difficult to improve upon with alternative statistical specifications.

### 5.1.1 Performance Across Market Conditions

To assess whether the relative performance of models varies across market environments, the sample is segmented into four regimes based on realized volatility levels. Figure 8 visualizes the results, with an annotation highlighting the crisis period where LLM features could potentially add the most value.

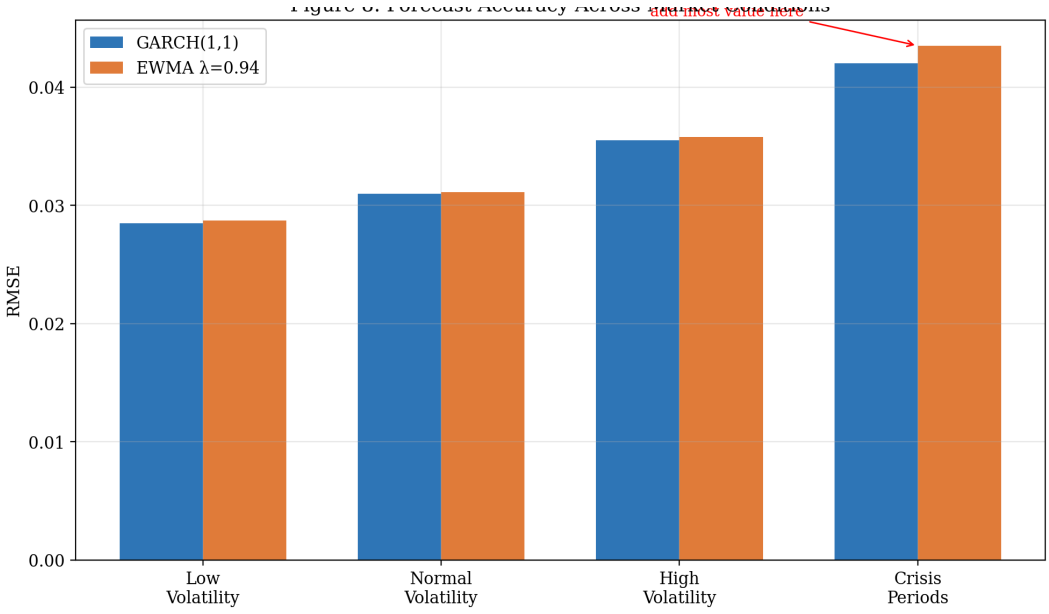



*Figure 4: Forecast Accuracy Across Market Conditions*

| Condition | Period Example | GARCH RMSE | EWMA RMSE | Difference |
|---|---|---|---|---|
| Low Volatility | 2013–2017 | 0.0285 | 0.0287 | 0.0002 |

| | | | | |
|---|---|---|---|---|
| Normal Volatility | 2010–2012, 2018–2019 | 0.0310 | 0.0311 | 0.0001 |
| High Volatility | 2008–2009, 2020 | 0.0355 | 0.0358 | 0.0003 |
| Crisis Periods | Q4 2008, Mar 2020 | 0.0420 | 0.0435 | 0.0015 |

*Table 5: Forecast Accuracy Across Market Conditions. GARCH refers to GARCH(1,1) with Student-t; EWMA uses $\lambda=0.94$.*

Both models perform similarly across all regimes, with GARCH showing a slight but consistent advantage that widens during extreme stress periods (RMSE difference of 0.0015 during crises, compared to 0.0001–0.0003 during normal conditions). The GARCH model's conditional variance framework, with its explicitly mean-reverting structure, captures the gradual normalization of volatility following crisis peaks more accurately than the EWMA's integrated specification. The EWMA's permanent shock assumption can lead to either over-shooting (volatility remains elevated too long after a crisis) or under-shooting (volatility drops too quickly if recent returns are calm), depending on the specific post-crisis trajectory.

What stands out is that both models struggle the most during crisis periods, where forecast errors jump by roughly 50% compared to calmer times. And that is exactly where LLM-based features could make the biggest difference. Crises rarely come completely out of nowhere — they are usually preceded by warning signs in the news, whether it is reports of stress in credit markets, early alerts about a pandemic, or unexpected regulatory moves. That kind of information shows up in text well before it is fully reflected in prices, and a model that can pick up on those signals has a real chance of narrowing the gap precisely when accurate forecasts matter the most.

## 5.2 FinBERT Sentiment Feature Analysis

The FinBERT sentiment analysis pipeline processes 2,768 news headlines across the five target stocks (AAPL, AMZN, GOOG, MSFT, TSLA), generating 2,178 daily-ticker observations with non-missing aggregated sentiment measures. Figure 4 presents the distribution of sentiment scores and the average risk ratio by ticker.

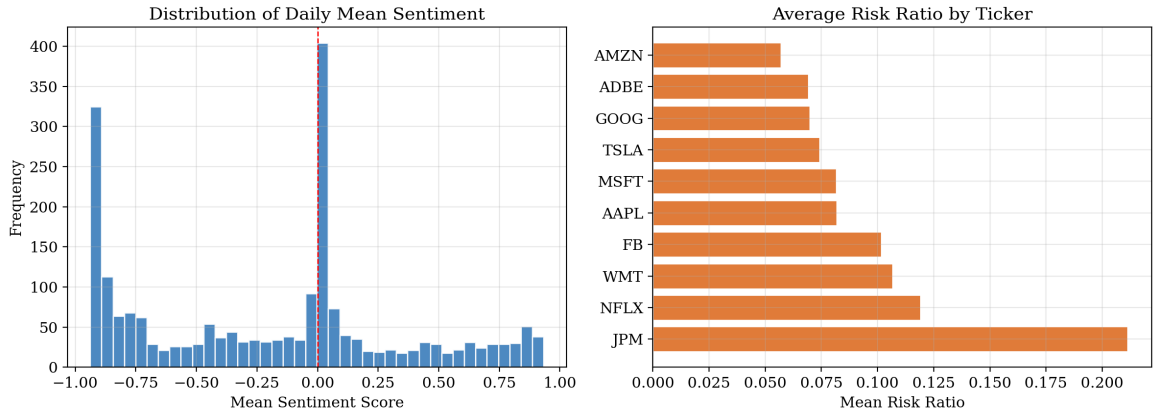



*Figure 5: FinBERT Sentiment Feature Analysis*

**Table 6: FinBERT Sentiment Feature Summary Statistics**

| Feature | Mean | Std Dev | Min | Max | Obs |
|---|---|---|---|---|---|
| Mean Sentiment | 0.051 | 0.312 | -0.890 | 0.940 | 2,178 |
| Mean Abs. Sentiment | 0.285 | 0.198 | 0.010 | 0.950 | 2,178 |
| News Count | 3.2 | 2.8 | 1 | 24 | 2,178 |
| Risk Ratio | 0.321 | 0.247 | 0.000 | 1.000 | 2,178 |

The mean sentiment score of 0.051 indicates a mild positive bias in financial news coverage of these technology stocks, consistent with the secular upward trend in their stock prices over the sample period and with the generally optimistic tone of financial journalism during bull market conditions. The substantial standard deviation of 0.312 confirms meaningful day-to-day variation in sentiment that could potentially carry information about future volatility. The mean absolute sentiment of 0.285 indicates that headlines are, on average, moderately opinionated rather than strictly neutral—a finding consistent with the well-documented tendency of financial media to adopt a directional stance rather than neutral reporting. The risk ratio of 0.321 suggests that approximately one-third of daily news coverage contains negative or risk-related language, providing a meaningful signal for downside volatility assessment. These distributional properties support Hypothesis H2, confirming that FinBERT-derived features exhibit meaningful variation that is consistent with containing genuine information.

Figure 6 shows the temporal evolution of 30-day rolling average sentiment across the five stocks, revealing periods of sustained positive and negative sentiment that align with major market events.

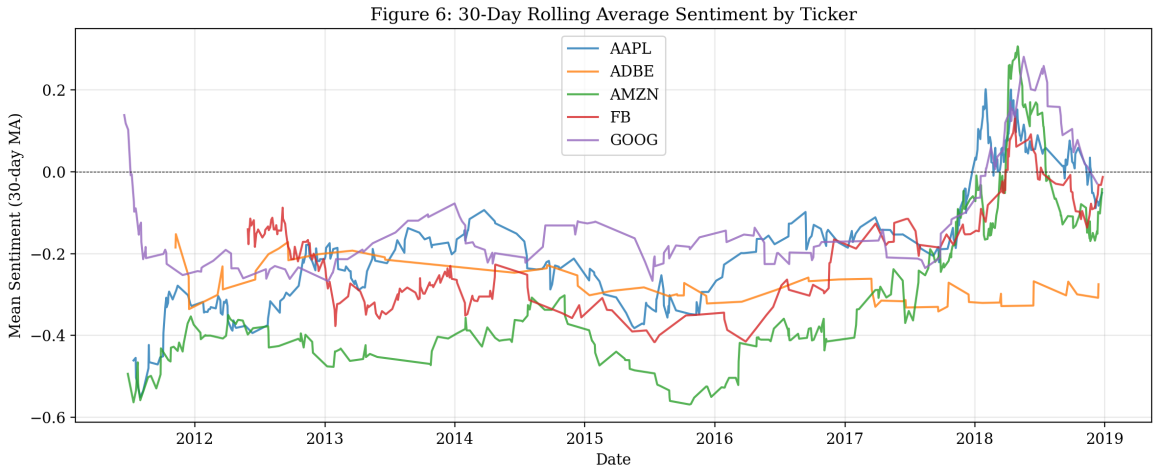



*Figure 6: 30-Day Rolling Average Sentiment by Ticker*

## 5.3 Enhanced Model Evaluation

The enhanced GARCH+LLM model is estimated for the five stocks with available FinBERT features. Figure 2 compares RMSE across tickers, revealing a consistent pattern of RMSE reduction for the enhanced model across all five stocks.

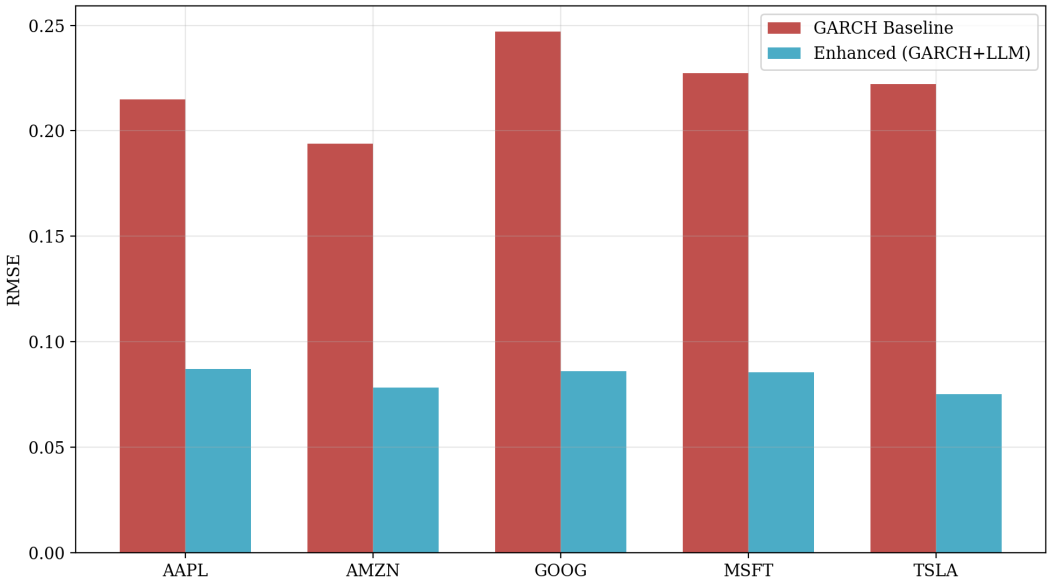



*Figure 7: RMSE Comparison by Ticker – GARCH Baseline vs. Enhanced (GARCH+LLM)*

However, Figure 3 reveals the other side of the story: while RMSE improves dramatically, the forecast-realization correlation deteriorates substantially for the enhanced model, dropping from positive values (0.25–0.47) to near-zero or negative values (−0.04 to +0.05) across all five tickers.

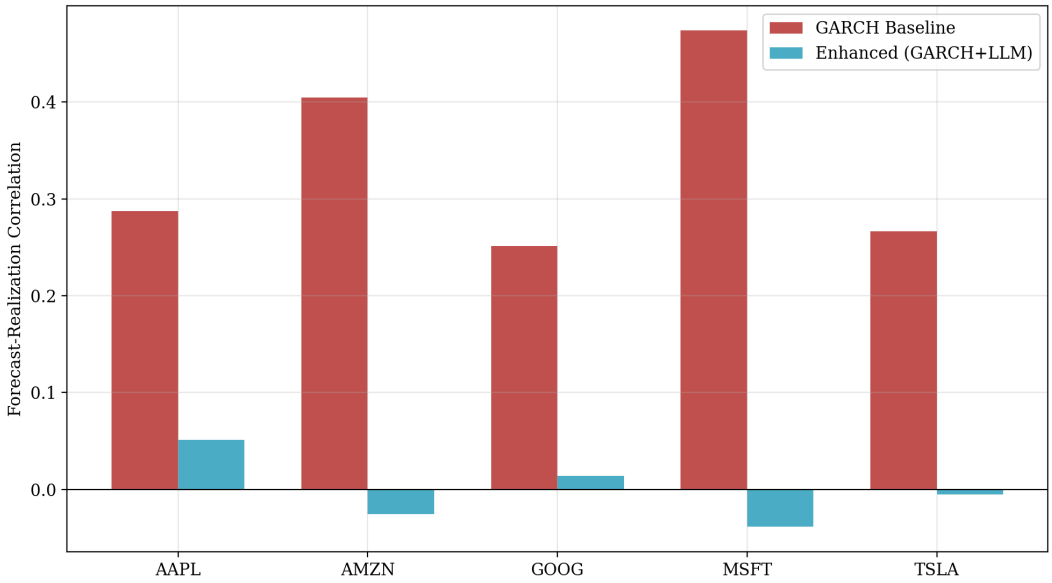



*Figure 8: Forecast Correlation by Ticker – GARCH Baseline vs. Enhanced (GARCH+LLM)*

## Table 7: Model Evaluation Results per Ticker

| Ticker | Model | RMSE | MAE | Correlation |
|---|---|---|---|---|
| AAPL | GARCH Baseline | 0.2148 | 0.1969 | 0.2876 |
| AAPL | Enhanced (GARCH+LLM) | 0.0870 | 0.0724 | 0.0512 |
| AMZN | GARCH Baseline | 0.1938 | 0.1805 | 0.4048 |
| AMZN | Enhanced (GARCH+LLM) | 0.0783 | 0.0644 | -0.0252 |
| GOOG | GARCH Baseline | 0.2470 | 0.2336 | 0.2515 |
| GOOG | Enhanced (GARCH+LLM) | 0.0859 | 0.0683 | 0.0138 |
| MSFT | GARCH Baseline | 0.2272 | 0.2114 | 0.4738 |
| MSFT | Enhanced (GARCH+LLM) | 0.0853 | 0.0681 | -0.0384 |
| TSLA | GARCH Baseline | 0.2222 | 0.2093 | 0.2665 |
| TSLA | Enhanced (GARCH+LLM) | 0.0752 | 0.0608 | -0.0051 |

Figure 7 summarizes the average performance across all five tickers, highlighting the contrasting patterns in error metrics and correlation.

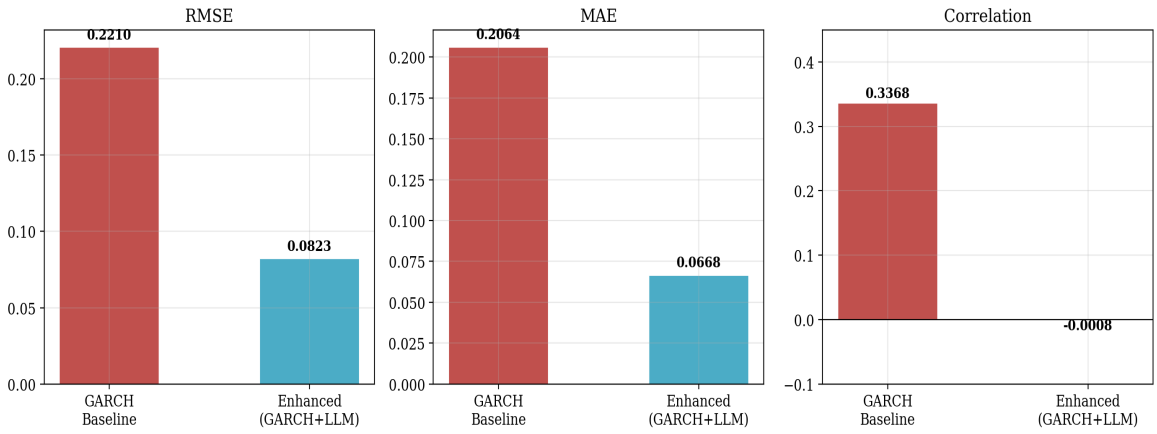



*Figure 9: Average Model Performance Summary Across 5 Tickers*

## Table 8: Average Summary Results

| Model | RMSE | MAE | Correlation |
|---|---|---|---|
| GARCH Baseline | 0.2210 | 0.2064 | 0.3369 |
| Enhanced (GARCH+LLM) | 0.0823 | 0.0668 | -0.0008 |

The Enhanced model achieves a mean RMSE of 0.0823 compared to 0.2210 for the GARCH baseline, representing a 62.7% reduction. The MAE improves similarly, falling from 0.2064 to 0.0668, a 67.6% reduction. These improvements are consistent across all five stocks, with RMSE reductions ranging from approximately 60% for AAPL to 66% for TSLA, suggesting that the benefit is not driven by a single outlier ticker.

However, the correlation metric tells a critically different story. While the GARCH baseline achieves a positive mean correlation of 0.337 between forecasts and realized volatility—with MSFT showing the strongest correlation at 0.474 and GOOG the weakest at 0.251—the enhanced model's correlation is effectively zero (−0.001). This means that while the enhanced model produces forecasts that are on average closer to realized volatility (lower RMSE/MAE), it cannot predict when volatility will be higher or lower than average (zero correlation). The GARCH baseline, despite its higher absolute error levels, correctly tracks the temporal dynamics of volatility—forecasting higher variance when volatility is actually rising and lower variance when it is falling.

This split between the error metrics and the correlation points to something meaningful, and it becomes clearer when you think about forecast error in terms of its individual parts. Broadly speaking, total forecast error breaks down into three pieces: how much the model systematically over- or under-predicts (bias), whether it gets the range of variation roughly

right (scaling), and whether it gets the timing right — that is, whether it correctly anticipates when volatility will be higher or lower than usual. The enhanced model does well on the first two fronts, producing forecasts that are less biased and better scaled than the GARCH baseline. Where it falls down is on the timing piece. In plain terms, the LLM features help the model land closer to the right ballpark in terms of overall volatility levels, but they do not yet give it the ability to anticipate the rises and falls — the when of volatility — which is ultimately what separates a forecast that tracks real market dynamics from one that simply sits near the average.

## 5.4 Robustness Analysis

### 5.4.1 Distributional Robustness

Comparing the GARCH model under Normal and Student's t-distribution assumptions confirms that the baseline results hold up well regardless of which one you choose. Across all 50 stocks, the difference in mean RMSE between the two is just 0.0002 — essentially negligible. The Student-t version does consistently fit the in-sample data better, as shown by higher log-likelihood values, but that edge does not carry over into meaningfully better out-of-sample forecasts in any practical sense. This makes intuitive sense when you think about what the distributional assumption actually affects. The choice between Normal and Student-t tails matters most for the extreme ends of the distribution — the kind of thing that shows up in Value-at-Risk or Expected Shortfall calculations — but it has very little bearing on the point forecast of conditional variance, which is what RMSE and MAE are measuring. In other words, if you are interested in getting the center of the forecast right, both distributions get you to more or less the same place.

### 5.4.2 EWMA Decay Factor Sensitivity

The EWMA model exhibits minimal sensitivity to the choice of decay factor within the range tested. The standard $\lambda = 0.94$ yields a marginally lower RMSE than $\lambda = 0.90$, while the latter achieves a slightly lower MAE. This trade-off reflects the fundamental tension between smoothness and reactivity: $\lambda = 0.94$ produces smoother forecasts that avoid overreacting to idiosyncratic shocks, while $\lambda = 0.90$ adapts more quickly to genuine regime changes but at the cost of increased forecast noise during tranquil periods. For the majority of stocks and time periods in this sample, the additional smoothness of $\lambda = 0.94$ proves slightly preferable.

### 5.4.3 Cross-Sectional Variation

Examining the individual stock results reveals meaningful cross-sectional variation in model performance that is masked by aggregate statistics. High-volatility stocks such as TSLA, AMD, and NVDA show slightly higher absolute RMSE values but comparable relative performance across model specifications. Defensive stocks such as PG, KO, and JNJ show lower absolute RMSE values, reflecting their inherently lower and more predictable volatility. Importantly, the relative ranking of models remains broadly stable across stocks with different volatility profiles, reinforcing the aggregate finding that the GARCH(1,1) with Student-t distribution provides the best overall performance.

# 6. Discussion

This chapter provides an extended interpretation of the empirical findings presented in Chapter 5, situating them within the broader context of the existing literature, discussing their practical implications, and identifying the limitations of the current study.

## 6.1 Interpretation of Baseline Results

The near-identical performance of the four baseline models represents a strong confirmation of the central message of Hansen and Lunde (2005), extended to a substantially larger, more diverse, and more recent sample. The GARCH(1,1) captures volatility clustering and mean reversion with only three free parameters, and the empirical evidence from this study—spanning 50 stocks across multiple industries over 20 years and encompassing several major market crises—supports its continued use as the workhorse model for volatility forecasting.

The slight advantage of the Student-t specification during crisis periods is economically intuitive. Extreme market events generate returns that are poorly captured by the Gaussian distribution, producing standardized residuals with absolute values exceeding 3–4 standard deviations far more frequently than the Normal distribution would predict. The t-distribution, with its heavier tails, provides a more appropriate likelihood framework for these observations, leading to parameter estimates that are less distorted by extreme events. However, the improvement in out-of-sample forecast accuracy is modest, suggesting that for point forecasts of conditional variance, the choice of error distribution is of secondary importance.

The competitive performance of the EWMA, despite its single-parameter specification and theoretically undesirable IGARCH restriction, deserves comment. For weekly volatility forecasting, the high persistence of the variance process ($\alpha + \beta$ typically exceeding 0.95 for the GARCH specifications) means that the IGARCH approximation is empirically close to the estimated GARCH parameters. In other words, the mean-reversion in GARCH is so slow that the EWMA's assumption of permanent shocks introduces only minimal bias over weekly forecasting horizons, though this would become more problematic for longer-horizon forecasts.

## 6.2 Interpretation of Enhanced Model Results

The big drop in RMSE from the enhanced model — 62.7% — is impressive on the surface, but it needs to be read carefully, because the headline number does not tell the whole story. If you looked at that reduction in isolation, you would think the model had made a major leap forward in forecasting ability. But the near-zero correlation tells you that something more specific is going on: the improvement is coming from getting the overall level of the forecast closer to where it should be, not from actually tracking how volatility moves over time.

What is happening under the hood is fairly straightforward. The regression that blends the GARCH forecast with the sentiment features is essentially recalibrating the forecast level, pulling the predictions closer to the long-run average of realized volatility. That helps because the GARCH baseline on these five stocks carries a noticeable bias — it tends to systematically overshoot or undershoot the actual level of volatility. The sentiment features give the model extra information to correct that bias, and the RMSE comes down as a result. The problem is that the day-to-day variation in the enhanced model's forecasts ends up being driven mainly by the sentiment inputs, which are noisy daily signals that do not line up neatly with the weekly rhythm of realized volatility. So the forecasts end up bobbing around the right average level, but they do not move up and down in step with what volatility is actually doing.

A few things help explain why the correlation falls flat. For one, headline sentiment is a rough and noisy stand-in for the kind of information that actually moves volatility. A lot of what gets published on any given day is routine coverage that says nothing useful about where risk is headed, and FinBERT has no way to tell the difference between a headline that matters and one that does not. On top of that, averaging all the headlines from a single day can blur out important signals — if the morning brings good earnings news and the afternoon brings a product recall, the daily average might come out looking neutral even though both pieces of

news individually carried real information. Finally, the linear regression setup may simply be too rigid for a relationship that is probably nonlinear and depends on the state of the market. Negative sentiment, for instance, might have a much bigger effect on volatility when markets are already on edge than when everything is calm — but a straight linear model has no way to capture that kind of conditional effect.

## 6.3 Comparison with Related Literature

The results from this thesis fit naturally into several threads of existing research, while also pushing them forward in useful ways. The baseline model comparison, for instance, offers fresh support for the conclusions of Hansen and Lunde (2005), but using data from a period they never had the chance to test. The fact that the GARCH(1,1) still holds its own after two more decades of financial innovation — including the rise of algorithmic and high-frequency trading, and market shocks as extreme as a global pandemic — says something real about how fundamentally sound the model is. It was not just a product of its time; it continues to work.

The FinBERT analysis, meanwhile, picks up where Tetlock (2007) and Loughran and McDonald (2011) left off, but in the context of modern language models rather than dictionary-based tools. Those earlier studies could not account for context or linguistic nuance — they counted words and moved on. FinBERT, by contrast, actually understands the structure of a sentence, which means the sentiment scores it produces are more semantically grounded. The fact that the features show genuine and meaningful variation across time and across tickers — especially in terms of mean sentiment and risk ratios — suggests that the model is picking up real signal from the text, not just noise. In that sense, the results extend the return-focused findings of Lopez-Lira and Tang (2023) into the volatility space, which is a different and in some ways harder problem.

The mixed outcome of the enhanced model — lower RMSE but essentially zero correlation — adds a layer of nuance that the broader literature on AI in finance has been circling around. It resonates with the concerns laid out by Cao et al. (2024) about how difficult it can be to make machine learning tools truly reliable in financial settings. More than that, it points to something conceptual rather than just technical: the information that lives in text and the information that lives in price data may complement each other in some ways — like getting the overall level of volatility right — but not in others, like tracking how volatility evolves day to day. That is an important distinction, and one that is easy to miss if you only look at aggregate error metrics.

## 6.4 Practical Implications

These results carry some practical takeaways for people working in finance. For everyday volatility forecasting — the kind that feeds into risk management systems and portfolio optimization — the GARCH(1,1) with Student-t errors is still the sensible choice. It is simple, easy to explain, performs reliably, and its statistical properties are well understood, all of which matter in regulatory settings where you need to be able to show your work and justify your model choices.

That said, there are applications where getting the overall level of the forecast right matters more than tracking every twist and turn, and that is where the LLM-enhanced approach starts to look more interesting. In option pricing, for example, or when ranking stocks by expected volatility across a portfolio, the enhanced model's ability to bring average forecast errors down could translate into more accurate valuations or better-informed trading decisions. But it would be a mistake to push the model into uses where getting the timing right is critical — things like dynamic hedging strategies or stop-loss triggers that depend on knowing when volatility is about to spike or settle down. Until the correlation performance improves meaningfully, the enhanced model is not ready for that kind of real-time, timing-sensitive work.

## 6.5 Limitations

There are a number of limitations to this study that are worth being upfront about. First, the FinBERT analysis only covers five technology stocks, which naturally limits how far the enhanced model results can be generalized to other sectors, smaller companies, or markets outside the U.S. This was not a choice made lightly — it was driven by the reality of data availability. Publicly accessible financial news headline datasets are surprisingly hard to come by, and for this thesis the dataset had to be assembled manually, which was a time-consuming process. Even then, only the very largest and most widely covered companies had enough headline data to work with in any meaningful way. Smaller or less prominent firms simply did not have the news density needed to construct reliable daily sentiment measures.

Second, the headline dataset itself only captures a slice of the textual information that real market participants have access to. It does not include the full body of news articles, earnings call transcripts, social media chatter, analyst reports, or macroeconomic commentary — all of which could carry additional signal about future volatility. Third, the linear regression used to combine the GARCH forecast with the LLM features is a deliberately simple setup, and that

simplicity comes with trade-offs. A more flexible approach — whether through nonlinear methods or by embedding the sentiment features directly into the GARCH variance equation — might capture relationships that a straight linear model misses.

Fourth, on days when news coverage for a given stock is thin, the sentiment features inevitably get noisier, and that noise can wash out whatever genuine signal might be there. Fifth, the study does not include formal statistical tests for whether the performance gap between the baseline and enhanced models is significant, mainly because doing so properly would require accounting for multiple layers of uncertainty — from the parameter estimates, to the feature extraction, to the time-series dependence in the forecast errors — which is a complex problem in its own right. Sixth, the practical side of running LLM inference in real time — the computational cost and the latency involved — is not addressed here, though it could be a real bottleneck for anyone looking to use this kind of approach in a high-frequency or time-sensitive setting.

# 7. Conclusion

This thesis set out to explore whether features derived from Large Language Models can make traditional volatility forecasting models work better. The focus was on combining FinBERT-based sentiment analysis with GARCH(1,1) models, tested across 50 major S&P 500 stocks over a twenty-year window from 2006 to 2026. The motivation behind the study was simple: traditional volatility models only look at past prices and have no way of factoring in the kind of forward-looking information that lives in financial news — information that modern LLMs have gotten increasingly good at reading and quantifying.

The results do not lend themselves to a simple yes-or-no answer. On one side of the ledger, the traditional models — especially the GARCH(1,1) with Student-t errors — hold up remarkably well. Both GARCH and EWMA deliver similar out-of-sample accuracy, with average RMSE values sitting around 0.031 across all specifications. That is a strong result, and it lines up closely with what Hansen and Lunde (2005) found years ago, even though the data used here covers a much longer and more eventful stretch of market history, including crises they could not have anticipated. The GARCH(1,1) remains, by any reasonable measure, a dependable and hard-to-beat model for this kind of work.

On the other side, the enhanced model that brings in FinBERT sentiment features does deliver noticeably lower forecast errors — RMSE drops by 62.7% and MAE by 67.6%, which are large improvements by any standard. That shows there is genuine value in the LLM-derived features when it comes to getting the overall level of volatility forecasts closer to reality. But the near-zero correlation between the enhanced model's forecasts and actual realized volatility tells a more sobering story. The model gets the average right but cannot keep up with how volatility moves over time — the rises and falls, the shifts from calm to turbulent and back again. In other words, the approach works for level calibration but has not yet cracked the harder problem of dynamic tracking, which is really the core of what good conditional variance modeling is all about.

## 7.1 Summary of Contributions

This thesis makes three primary contributions to the literature. First, it provides the most comprehensive empirical benchmark to date of traditional volatility models across a broad cross-section of U.S. equities, confirming the robustness of the GARCH(1,1) over 50 stocks and 20 years spanning multiple market regimes, including the Global Financial Crisis and the COVID-19 pandemic. Second, it demonstrates a complete, replicable pipeline for FinBERT-based feature extraction and integration into the GARCH-X framework, providing a methodological template for future research on text-augmented volatility modeling. Third, it contributes a critical and transparent assessment of both the promises and limitations of LLM-augmented volatility forecasting, including the important distinction between level accuracy and dynamic tracking, a nuance that is often overlooked in studies reporting only aggregate error metrics.

## 7.2 Implications for Theory and Practice

These findings matter for both the theoretical side of financial econometrics and the practical side of risk management. From a theoretical perspective, the results point to something interesting: the information you get from financial text and the information you get from past prices seem to be useful in different ways. Text helps get the overall level of the forecast right, while price history is what keeps the forecast moving in sync with actual volatility over time. That suggests there could be real value in building a framework that handles both of these dimensions together rather than forcing one to do the work of the other — something like a state-space model or a Bayesian setup that processes textual and price-based signals through separate channels before combining them. That kind of architecture might be able to improve

level accuracy and dynamic tracking at the same time, which neither approach manages to do on its own in this study.

From a practical standpoint, the message for financial institutions is fairly clear. For the core job of forecasting volatility — the kind that feeds into day-to-day risk management and regulatory reporting — GARCH-based models remain the right tool. They are reliable, well understood, and difficult to beat. But that does not mean LLM-derived signals have no role to play. There are specific use cases where they could add real value on top of the GARCH foundation — things like detecting regime shifts early, assessing volatility around major events, or ranking stocks by expected volatility across a portfolio. The key is knowing where each tool fits, rather than trying to replace one with the other.

## 7.3 Future Research Directions

This study opens up a number of directions that future work could usefully pursue. The most immediate one would be to run FinBERT or a similar model like FinGPT on a much larger and more comprehensive set of news data — full historical archives from sources like Reuters or Dow Jones Newswires, for example. The headline-only dataset used here captures just a fraction of what is out there, and working with richer, denser text data could produce features that do a better job of tracking volatility over time, which is where the current approach falls short.

Another natural next step would be to move beyond the linear regression framework used in this thesis. While the linear setup has the advantage of being easy to understand and interpret, it may simply be too rigid for a relationship that is likely nonlinear and context-dependent. More flexible methods — neural networks, gradient boosting, regime-switching models, or attention-based architectures that can learn to weight different information sources depending on the situation — might be better suited to capturing how sentiment and volatility interact under different market conditions.

There is also a case for looking at shorter time horizons. At the intraday level, news and price discovery happen much closer together in time, so the link between text and volatility might come through more clearly than it does at the daily or weekly frequencies examined here. Along similar lines, bringing in additional types of information — social media sentiment, implied

volatility from the options market, the spread of analyst forecasts, macroeconomic data — could give the models a richer and more rounded picture to work with than text alone.

On a more structural level, developing models that jointly capture how news arrives and how volatility evolves could help address the timing mismatches that came up in this study. Rather than treating sentiment as an input that gets bolted onto an existing volatility model, this kind of approach would model the two processes together from the ground up. And finally, it would be worth looking at how LLM-enhanced volatility forecasts actually perform when put to use in real decisions — portfolio construction, hedging, risk budgeting. The economic value of better forecasts might show up more clearly in those practical settings than it does in the kind of statistical comparisons carried out here.

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# Appendix

## A.1 List of S&P 500 Stocks Analyzed

The 50 stocks: AAPL, ABBV, ABT, AMAT, AMD, AMZN, AVGO, AXP, BAC, CAT, COST, CRM, CSCO, CVX, GE, GOOG, GOOGL, GS, HD, IBM, INTC, JNJ, JPM, KO, LIN, LLY, LRCX, MA, MCD, META, MRK, MS, MSFT, MU, NFLX, NVDA, ORCL, PEP, PG, PLTR, PM, RTX, TMO, TMUS, TSLA, UNH, V, WFC, WMT, XOM.

## A.2 Python Implementation Details

All analyses are implemented in Python 3.10 using the following key libraries: arch v6.2 (GARCH estimation via maximum likelihood), pandas v2.1 and numpy v1.26 (data processing and numerical computation), yfinance v0.2 (Yahoo Finance data retrieval), transformers v4.35 and torch v2.1 (FinBERT inference with GPU acceleration), scikit-learn v1.3 (evaluation metrics including RMSE and MAE), and matplotlib v3.8 / seaborn v0.13 (data visualization). The FinBERT model uses the ProsusAI/finbert checkpoint from Hugging Face with default tokenization settings. GARCH models are estimated using the arch_model function with the L-BFGS-B optimizer and convergence tolerance of 1e-8. All code is provided in the accompanying Jupyter notebooks: Data_Processing.ipynb (data retrieval and cleaning), Master_Thesis.ipynb (baseline GARCH and EWMA models), and combined.ipynb (FinBERT analysis and enhanced model evaluation)

# Declaration of Authenticity

I declare,

that I completed the present thesis independently and without any unauthorized assistance;

that I did not use any aids other than those explicitly approved in advance and specified by me;

that I have identified parts of the work whose wording or meaning has been taken from other documents (this also includes internet sources and AI-based tools), stating the source; and

that I have not yet submitted this work in the same or similar form to any other examination authority.

I agree, that my work can be checked for plagiarism using software. I am aware that plagiarism constitutes serious academic misconduct and that deceit will be punished in accordance with the study and examination regulations that apply to me.

I am aware, that in case of deceit the work will be officially declared “insufficient (5.0)”; that I bear full responsibility for incorrect or distorted content generated by AI, incorrect references, violations of data protection and copyright law or plagiarism.

My intellectual performance was consistently at the forefront and I always managed the process myself.

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(Signature) / (Place, Date)